



ITAHARI
INTERNATIONAL
COLLEGE



LONDON
METROPOLITAN
UNIVERSITY

Module Code & Module Title

CC5051NT Databases

Assessment Type

Coursework

Semester

2024/25 Autumn

Student Name: Anisha Karki

London Met ID: 23049162

College ID: NP05CP4S230070

Assignment Due Date: Thursday, January 23, 2025, 2025

Assignment Submission Date: Thursday, January 23, 2025

Submitted To: Ajayraj Bhattarai

Word Count (Where Required):6088

I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Table of Contents

1.	Introduction.....	1
1.1.	Relational Databases.....	1
1.2.	Oracle SQL PLUS.....	1
1.3.	Introduction of the business and its forte.....	1
1.4.	Current Business Activities and Operations.....	3
1.5.	Business Rules.....	4
2.	Initial ERD.....	4
2.1.	Identification of Entities and Attributes.....	4
2.2.	Identification and Representation of the Primary Keys.....	7
2.3.	Relationships:.....	8
2.4.	Improved ERD Text Description.....	8
2.5.	Entity Relationship Diagram (ERD).....	12
3.	Normalization.....	13
3.1.	UNF.....	14
3.2.	1NF (First Normal Form).....	15
3.3.	2NF (Second Normal Form).....	17
3.4.	3NF (Third Normal Form).....	22
4.	Data Dictionary.....	25
4.1.	STUDENT.....	26
4.2.	PROGRAM.....	26
4.3.	MODULE.....	27
4.4.	TEACHER.....	27
4.5.	RESOURCE.....	28
4.6.	ANNOUNCEMENT.....	28

4.7.	ASSESSMENT.....	29
4.8.	RESULT.....	29
4.9.	GRADE_FEEDBACK.....	30
4.10.	Program_Module (Linking Table).....	30
4.11.	Module_Teacher (Linking Table).....	31
5.	Final ERD (Entity-Relationship Diagram).....	31
6.	Implementation.....	32
7.	Database Querying.....	50
7.1.	Information query.....	50
7.2.	Transaction query.....	53
8.	Critical Evaluation.....	57
8.1.	Critical Evaluation of Module.....	57
8.2.	Critical Assessment of Coursework.....	59
9.	Drop Query and Database Dump file creation.....	61

Table of Figures

Figure 1 Student to Program.....	8
Figure 2 Program to Module.....	9
Figure 3 Initial ERD.....	12
Figure 4 Final ERD.....	31
Figure 5 Program table created.....	32
Figure 6 Student table created.....	32
Figure 7 Module table created.....	33
Figure 8 Teacher Table created.....	33
Figure 9 Resource table created.....	33
Figure 10 Announcement table created.....	34
Figure 11 Assessment table created.....	34
Figure 12 Result table created.....	35
Figure 13 Grade_Feedback table created.....	35
Figure 14 Program_Module table created.....	35
Figure 15 Module_Teacher table created.....	36
Figure 16 Detail inserted in Program.....	36
Figure 17 Detail inserted in Student.....	36
Figure 18 Detail inserted in Module.....	37
Figure 19 Detail inserted in Teacher.....	37
Figure 20 Detail inserted in "Resource".....	38
Figure 21 Detail inserted in Announcement.....	38
Figure 22 Detail inserted in Assessment.....	39
Figure 23 Detail inserted in Result.....	39
Figure 24 Detail inserted in Grade_Feedback.....	40
Figure 25 Detail inserted in Program_Module.....	40
Figure 26 Detail inserted in Module_Teacher.....	40
Figure 27 Details of Program.....	41
Figure 28 Details of Student.....	42
Figure 29 Details of Module.....	43
Figure 30 Details of Teacher.....	44
Figure 31 Details of "Resource".....	45
Figure 32 Details of Announcement.....	46
Figure 33 Details of Assessment.....	47
Figure 34 Details of Result.....	48

Figure 35 Details of Grade_Feedback.....	48
Figure 36 Details of Program_Module.....	49
Figure 37 Details of Module_Teacher.....	49
Figure 38 Program and total no. of student.....	50
Figure 39 Announcement.....	51
Figure 40 Module with D.....	51
Figure 41 not submitted.....	52
Figure 42 more than 1 module.....	52
Figure 43 Latest assessment deadline.....	53
Figure 44 Highest score of 3 student.....	54
Figure 45 Average in database.....	55
Figure 46 total no. and average in assessment.....	56
Figure 47 pass or fail in module.....	57
Figure 48 Drop Query.....	62
Figure 49 Data dump file creation.....	62
Figure 50 Expdp Anishakarki23049162.....	63

Table of Tables

Table 1 Student data dictionary.....	6
Table 2 Program data dictionary.....	6
Table 3 Module data dictionary.....	7
Table 4 Teacher data dictionary.....	7
Table 5 Resource data dictionary.....	7
Table 6 Announcement data dictionary.....	8
Table 7 Assessment data dictionary.....	8
Table 8 Result data dictionary.....	9
Table 9 Grade_feedback data dictionary.....	9
Table 10 Program_Module data dictionary.....	10
Table 11 Module_Teacher data dictionary.....	10

CC5051NT Databases

1. Introduction

1.1. Relational Databases

A relational database is a type of database that uses a structured tabular format to store data. The main and foreign keys are used by the table to predefine a relationship. In order to successfully extract data from these kinds of databases, query languages such as SQL-Structured Query Language are utilized.

1.2. Oracle SQL PLUS

Oracle SQL PLUS SQL*Plus is a command-line interface for Oracle SQL that connects to the Oracle database using a host language. This tool is used to execute SQL commands for creating databases, manipulating data, and querying.

1.3. Introduction of the business and its forte

Ms. Mary's E-Classroom Platform marks a notable improvement in the field of online education. It creates a well-organized and comprehensive learning space specially designed for college students. The platform's main goal is to improve the educational experience for both teachers and students by incorporating essential features such as student management, program organization, and module administration.

The platform's course management tools allow teachers to set up their classes with ease. Instructors can oversee the learning process and make adjustments as needed. Students can move through their lessons at a pace that suits them. This flexibility helps ensure that they fully understand each topic before advancing to the next one. The platform includes support for various academic tasks, such as

CC5051NT Databases

tracking assessments, which helps teachers monitor student progress efficiently. By providing clear performance metrics, teachers can assess how well students are doing and offer timely feedback.

Another vital aspect of the platform is its resource allocation feature. Teachers can distribute educational materials and assignments effectively, ensuring that every student has access to what they need for their studies. This organized distribution of resources enhances the learning experience. Additionally, the platform can generate reports that summarize student performance and course outcomes. These reports offer valuable insights that assist educators in adjusting their teaching methods to better meet the needs of their students.

The integration of technology is crucial for making learning easier and more reachable. The platform allows students to access materials in a structured way, creating a clear path through their learning journey. This organization helps students navigate their courses without confusion. Built-in communication tools promote interaction between students and teachers. These tools make it simple for students to ask questions and seek assistance when needed.

With these features, learning on Ms. Mary's E-Classroom Platform becomes engaging and interactive. Students are encouraged to stay motivated while moving through their studies in a lively setting. The platform not only supports academic achievement, but it also equips students with skills they will need for future challenges in their educational pursuits. Overall, Ms. Mary's E-Classroom Platform is a valuable tool for enhancing online education and fostering student success.

CC5051NT Databases

1.4. Current Business Activities and Operations

The college provides a wide range of courses in fields such as BSc in computing, networking, and multimedia, among others. Each of these programs is made up of various modules that cover different topics essential to the curriculum. Students will focus on one program at a time, allowing them to concentrate on their studies. In this setting, teachers play a crucial role in managing each module. They will post important announcements, create and manage various assessments, and keep track of student performance throughout the course.

The modules are thoughtfully designed with structured resources that guide students through the learning process. These resources include readings, videos, and assignments, all aimed at enhancing the educational experience. Teachers organize these resources to ensure that students complete them in a specific order. This structured approach helps to maintain a clear learning path.

Assessments are an essential part of each module. They are designed to evaluate students' understanding and knowledge of the material. After students take these assessments, teachers will produce results, which help both students and educators measure progress and identify areas that may need improvement.

In addition to assessments, teachers will also post notices related to the modules. These notices may include updates about the course schedule, reminders about upcoming deadlines, and additional resources to support student learning. Overall, the college creates an interactive educational environment where students receive guidance and support to succeed in their studies.

CC5051NT Databases

1.5. Business Rules

- a) One student can enroll in only one program.
- b) One program contains many mandatory modules.
- c) One module can be part of many programs.
- d) One module is taught by one or more teachers.
- e) Every module will have an ID and will be made up of resources and assessments.
- f) Each assessment is part of one and only one module. The attributes for the assessments are ID, title, deadline and weightage.
- g) Students will view the resources in the same order.
- h) Teachers can post announcements related to their respective modules.
- i) Each announcement belongs to one and only one module.
- j) Student results should be displayed to show total marks and also the breakdown of each module.

2. Initial ERD

The ERD described herein will form the basis for the development of the schema to ensure that all entities and their relationships will be accurately represented, supporting E-classroom functionalities. Relationship schemes support assurance, hereby termed interaction between students, teachers, programs, modules, assessments, resources, and announcement attributes of a database with a structured, dynamic educational environment in view.

2.1. Identification of Entities and Attributes

Before changing

CC5051NT Databases

1. Student (StudentID(PK), FirstName, LastName, Email, EnrollmentDate)
2. Program (ProgramID(PK), ProgramName, Description, Duration, TotalCredits)
3. Module (ModuleID(PK), ModuleName, Description, Credits, Semester)
4. Teacher (TeacherID(PK), FirstName, LastName, Email, Specialization)
5. Assessment (AssessmentID(PK), Title, Deadline, Weightage, MaximumMarks)
6. Result (ResultID(PK), ObtainedMarks, TotalMarks, Grade, Feedback)
7. Resource (ResourceID(PK), Title, Type, Duration, SequenceNumber)
8. Announcement (AnnouncementID(PK), Title, Content, PostDate, ExpiryDate)

After changing

1. Student
StudentID(PK)
FirstName
LastName
Email
EnrollmentDate
2. Program
ProgramID(PK)
ProgramName
Description
Duration
TotalCredits
3. Module
ModuleID(PK)

CC5051NT Databases

ModuleName

Description

Credits

Semester

TeacherID

FirstName

LastName

Email

Specialization

ResourceID

Title

Type

Duration

SequenceNumber

AnnouncementID

Title

Content

PostDate

ExpiryDate

AssessmentID

Title

Deadline

Weightage

MaximumMarks

ResultID

CC5051NT Databases

ObtainedMarks

TotalMarks

Grade

Feedback

2.2. Identification and Representation of the Primary Keys

Before changing

The primary keys for each entity have been identified as follows:

1. **Student: StudentID** (Primary Key)
2. **Program: ProgramID** (Primary Key)
3. **Module: ModuleID** (Primary Key)
4. **Teacher: TeacherID** (Primary Key)
5. **Assessment: AssessmentID** (Primary Key)
6. **Result: ResultID** (Primary Key)
7. **Resource: ResourceID** (Primary Key)
8. **Announcement: AnnouncementID** (Primary Key)

After changing

The primary keys for each entity have been identified as follows:

1. **Student: StudentID** (Primary Key)
2. **Program: ProgramID** (Primary Key)
3. **Module: ModuleID** (Primary Key)

2.3. Relationships:

1. One student is enrolled in one program, but one program can have several students.

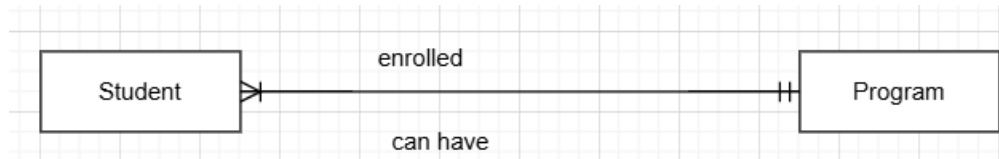


Figure 1 Student to Program

2. Every program consists of several modules, but a module can belong to several programs.

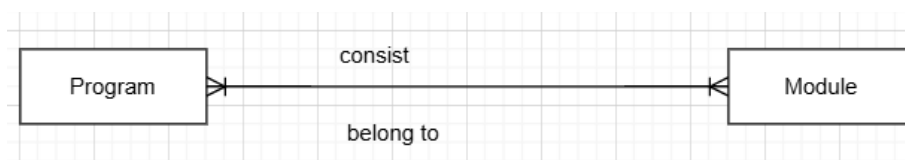


Figure 2 Program to Module

2.4. Improved ERD Text Description

Entities and Attributes

1. Student

- o StudentID (PK): Unique identifier for each student.
- o FirstName: Student's first name.
- o LastName: Student's last name.
- o Email: Student's email address.
- o EnrollmentDate: Date of enrollment

CC5051NT Databases

- ProgramID (FK): Links the student to a program.

2. Program

- ProgramID (PK): Unique identifier for each program.
- ProgramName: Name of the program.
- Description: Description of the program.
- Duration: Estimated of how long it will take to complete
- TotalCredits: Number of credits the module provides.

3. Module

- ModuleID (PK): Unique identifier for each module.
- ModuleName: Name of the module.
- Credits: Number of credits the module provides.
- Semester: a division of the academic year
- ProgramID (FK): Links the module to a program.
- TeacherID (PK): Unique identifier for each teacher.
- FirstName: First name of the teacher.
- LastName: Last name of the teacher.
- Email: Primary email address of the teacher.
- Specialization: experts in that specific field
- ResourceID (PK): Unique identifier for each resource.

CC5051NT Databases

- o Title: Title of the resource.
- o Type: Type of resource such as video, document.
- o Duration: Estimated of how long it will take to complete
- o ModuleID (FK): relates the resource to a module.
- o SequenceNumber: Number in which the resources must be completed.
- o AnnouncementID (PK): Unique identifier for each announcement.
- o Title: Title of the announcement.
- o Content: Content of the announcement.
- o PostDate: Date the announcement was posted.
- o ExpiryDate: Date the announcement will be removed.
- o ModuleID (FK): relates the announcement to a module.
- o AssessmentID (PK): Unique identifier for each assessment.
- o Title: Title of the assessment.
- o Deadline: Date for submission.
- o Weightage: Percentage weight of the assessment.
- o MaximumMarks: Highest marks
- o ModuleID (FK): The assessment for a module.
- o ResultID (PK): Unique identifier for each result entry.

CC5051NT Databases

- StudentID (FK): Result belongs to a student.
- AssessmentID (FK): Links the result to an assessment.
- ObtainedMarks: Marks the student achieved in the assessment.
- TotalMarks: Number of credits the module provides.
- Grade: final marks awarded to a student based on their performance in assessments, exams, or coursework.
- Feedback: Feedback by the teacher.

2.5. Entity Relationship Diagram (ERD)

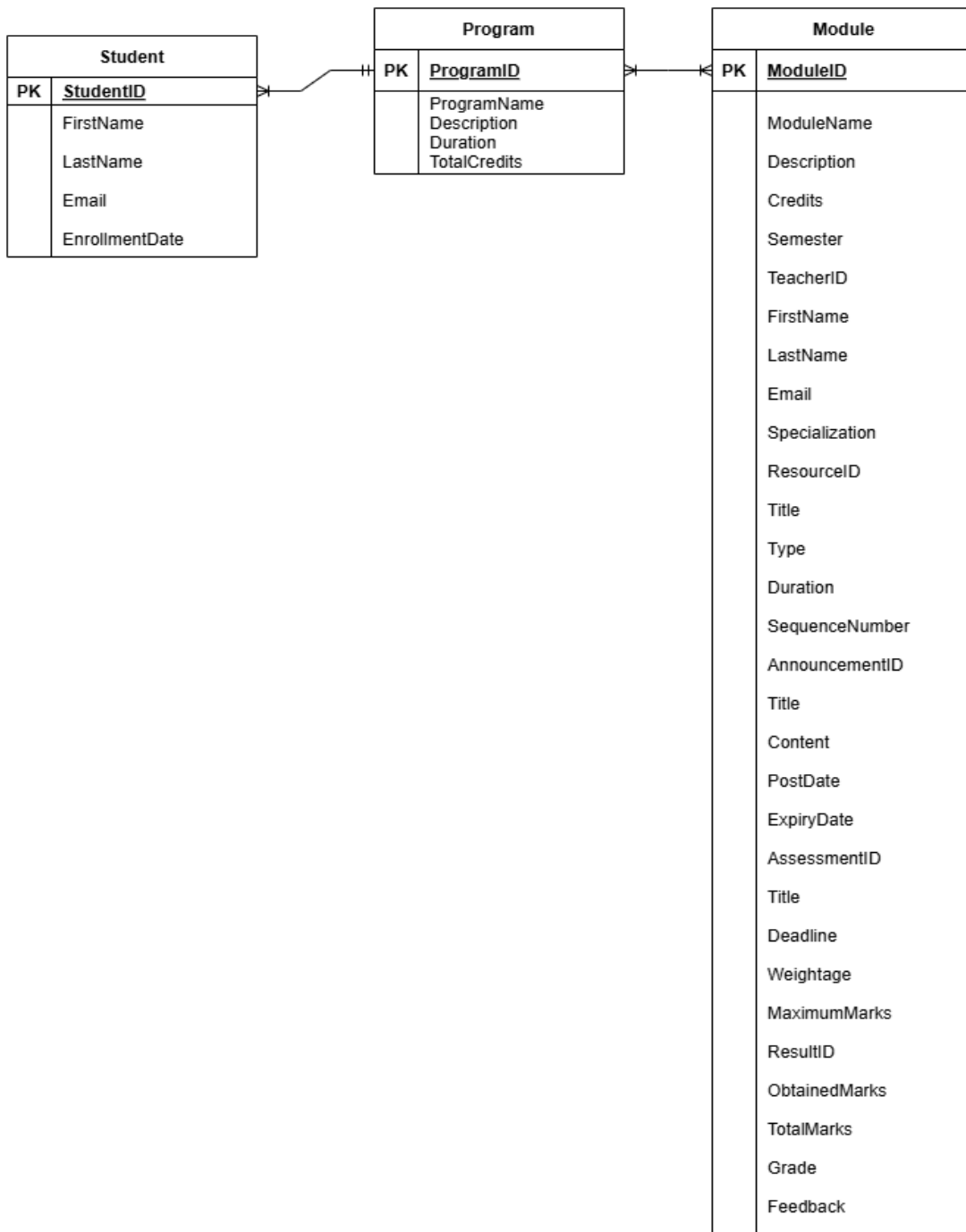


Figure 3 Initial ERD

3. Normalization

Normalization is an important method used in database design that aims to arrange data in a clear and efficient way. The main purpose of normalization is to simplify complex data structures. This is done by taking a large, complicated table and breaking it down into smaller, easier-to-handle tables. At the same time, it is vital to keep the relationships between these tables meaningful and coherent.

There are several reasons why normalization is used in database design. First, it helps reduce data redundancy. When data is duplicated across multiple tables, it takes up unnecessary space and can create inconsistencies. By removing duplicate data, normalization not only saves storage space but also improves the overall quality of the data.

Second, normalization enhances data integrity. A well-structured database minimizes the chances of errors during data entry, updates, and deletions. This means that the information contained in the database is more accurate and consistent, reducing the risk of anomalies that can affect the reliability of the data.

Third, normalization increases data flexibility. Databases that are organized according to normalization principles can be changed and adapted more easily. This flexibility is crucial for responding to new requirements or updating existing structures without significant difficulties.

Lastly, normalization simplifies the process of data retrieval. When tables are organized logically and relationships are clear, it becomes easier to query the database for specific information. Users can quickly find the exact data they need without sifting through unnecessary information.

CC5051NT Databases

The process of normalization involves following a series of steps known as Normal Forms. These steps guide database designers in ensuring that their tables are structured correctly and effectively.

In summary, normalization is a key concept in database design that is vital for maintaining data quality and organization. By applying normalization techniques, designers can build databases that are efficient, consistent, and capable of meeting the ever-changing needs of applications. This leads to better performance and reliability in managing data over time.

3.1. UNF

Unnormalized Form (UNF)

In database normalization, the Unnormalized Form (UNF), also known as 0NF, represents the initial, raw state of data before any normalization rules are applied. It's characterized by:

- **Repeating Groups:** The presence of repeating groups of data within a single row. This means that multiple occurrences of the same set of attributes are associated with a single primary key value.

student(StudentID, FirstName, LastName, Email, EnrollmentDate, {ProgramID, ProgramName, Description, Duration, TotalCredits, {ModuleID, ModuleName, Description, Credits, Semester, {TeacherID, FirstName, LastName, Email, Specialization}, {ResourceID, Title, Type, Duration, SequenceNumber}, {AnnouncementID, Title, Content, PostDate, ExpiryDate}, {AssessmentID,

CC5051NT Databases

Title, Deadline, Weightage, MaximumMarks, {ResultID, ObtainedMarks, TotalMarks, Grade, Feedback} } })

3.2. 1NF (First Normal Form)

First Normal Form (1NF) is a key concept in database normalization. It represents the most basic level of database organization. A table is considered to meet the requirements of 1NF when it meets certain criteria.

The first requirement is that all values in a cell must be atomic. This means each cell should hold only a single, indivisible value. For instance, if a cell is meant to store a student's name, it should only contain that student's name, with no additional information or multiple names included.

The second requirement is that there should be no repeating groups in the table. This means fields should not contain multiple values or lists. For example, in a table listing students and their courses, a student should not have a cell that lists several courses together. Instead, each course should be represented in its own row or separate entry.

1NF serves as the foundation for database normalization. It is the first step in a process that leads to higher normal forms, such as Second Normal Form (2NF) and Third Normal Form (3NF). These higher levels build upon the basic principles established in 1NF.

One major benefit of 1NF is that it eliminates data redundancy. By ensuring that each value is atomic and that there are no repeating groups, 1NF helps to cut

CC5051NT Databases

down on duplicate data. This not only saves storage space but also enhances the integrity of the data.

Another key point is that 1NF simplifies data manipulation. When a table is structured according to 1NF, it becomes easier to insert, delete, or update data. This structured approach reduces the chances of causing inconsistencies or anomalies when making changes to the data.

Additionally, tables that comply with 1NF enable efficient queries. When data is organized in a clear and logical way, it is simpler for users to retrieve and analyze information. The more straightforward the structure, the quicker and easier the queries can be.

To transform a database into 1NF, the existing nested structures are broken down into multiple tables. Each attribute in these tables holds atomic values. For example, if a table includes modules, teachers, resources, and assessments, these repeating groups are split into their own distinct tables. This ensures each piece of information is clearly defined and organized, making the database more effective and easier to use.

Resulting Tables in 1NF:

1. STUDENT:

- StudentID (PK), FirstName, LastName, Email, EnrollmentDate, ProgramID (FK)

2. PROGRAM:

- ProgramID (PK), ProgramName, Description, Duration, TotalCredits

3. MODULE:

CC5051NT Databases

- ModuleID (PK), ModuleName, Description, Credits, Semester, ProgramID (FK)

4. **TEACHER:**

- TeacherID (PK), FirstName, LastName, Email, Specialization

5. **RESOURCE:**

- ResourceID (PK), Title, Type, Duration, SequenceNumber, ModuleID (FK)

6. **ANNOUNCEMENT:**

- AnnouncementID (PK), Title, Content, PostDate, ExpiryDate, ModuleID (FK)

7. **ASSESSMENT:**

- AssessmentID (PK), Title, Deadline, Weightage, MaximumMarks, ModuleID (FK)

8. **RESULT:**

- ResultID (PK), AssessmentID (FK), StudentID (FK), ObtainedMarks, TotalMarks, Grade, Feedback

9. **Linking Tables:**

- **Program_Module:** ProgramID (FK), ModuleID (FK)
- **Module_Teacher:** ModuleID (FK), TeacherID (FK)

3.3. **2NF (Second Normal Form)**

he foundation set by First Normal Form (1NF) and focuses on reducing redundancy in data. The main goal of 2NF is to remove partial dependencies.

CC5051NT Databases

This means that every non-key attribute must rely entirely on the entire primary key, not just a part of it.

To qualify for 2NF, a table must already be in 1NF. This means it should have all entries organized in such a way that each value is atomic, and there are no repeating groups. Once a table meets this first requirement, the next step is to eliminate Partial Functional Dependencies (PFD). Non-key attributes must be fully dependent on the complete composite primary key.

A partial dependency is when a non-key attribute only relies on a portion of the primary key. This can create issues such as data redundancy, where the same information is stored in multiple places, and anomalies during data updates or deletions.

2NF primarily addresses tables that have composite primary keys, which consist of two or more attributes that combine to form a unique identifier. For example, in a table where both StudentID and AssessmentID serve as the primary key, it is crucial for all non-key attributes to depend on both these identifiers together, rather than just one.

To be in 2NF, there are specific conditions that must be satisfied:

1. The table must first meet the requirements of 1NF.
2. There must be no Partial Dependencies present; all non-key attributes should depend on the full composite primary key.

Understanding functional dependencies is key in this process. For example, consider several entities within a database.

Functional Dependency Analysis (FFD and PFD):

1. **STUDENT:**

CC5051NT Databases

For the STUDENT table, the functional dependency shows that StudentID determines attributes such as FirstName, LastName, Email, EnrollmentDate, and ProgramID. In this case, there is no partial dependency since all attributes depend on StudentID.

- **FFD:** StudentID $\rightarrow$ {FirstName, LastName, Email, EnrollmentDate, ProgramID}.
- No PFD, as all attributes depend on StudentID.

2. PROGRAM:

Looking at the PROGRAM table, ProgramID serves as the unique identifier, with other attributes like ProgramName, Description, Duration, and TotalCredits fully depending on it. Again, there are no partial dependencies.

- **FFD:** ProgramID $\rightarrow$ {ProgramName, Description, Duration, TotalCredits}.
- No PFD, as all attributes depend on ProgramID.

3. MODULE:

In the MODULE table, ModuleID identifies attributes such as ModuleName, Description, Credits, Semester, and ProgramID, all of which are fully dependent on ModuleID, confirming no partial dependencies.

- **FFD:** ModuleID $\rightarrow$ {ModuleName, Description, Credits, Semester, ProgramID}.
- No PFD, as all attributes depend on ModuleID.

4. TEACHER:

CC5051NT Databases

The TEACHER table follows the same pattern where TeacherID is the unique identifier for attributes including FirstName, LastName, Email, and Specialization. These attributes are entirely dependent on TeacherID.

- **FFD:** TeacherID $\rightarrow$ {FirstName, LastName, Email, Specialization}.
- No PFD, as all attributes depend on TeacherID.

5. ASSESSMENT:

For ASSESSMENT, AssessmentID identifies Title, Deadline, Weightage, MaximumMarks, confirming that there are no partial dependencies.

- **FFD:** AssessmentID $\rightarrow$ {Title, Deadline, Weightage, MaximumMarks, ModuleID}.
- No PFD, as all attributes depend on AssessmentID.

6. RESULT:

In the RESULT table, however, the scenario changes. The functional dependency shows that both StudentID and AssessmentID together determine ObtainedMarks, TotalMarks, Grade, and Feedback. Here, a partial dependency occurs because the attributes Grade and Feedback depend solely on AssessmentID, not on the composite key formed by StudentID and AssessmentID. This indicates a need for adjustment.

- **FFD:** {StudentID, AssessmentID} $\rightarrow$ {ObtainedMarks, TotalMarks, Grade, Feedback}.
- **PFD:** AssessmentID $\rightarrow$ {Grade, Feedback}: These attributes depend only on AssessmentID, not the composite key {StudentID, AssessmentID}.

CC5051NT Databases

2NF Adjustments:

To comply with 2NF, adjustments must be made. The RESULT table can be split into two new tables. The first would continue to hold the result information such as ResultID, StudentID, AssessmentID, ObtainedMarks, and TotalMarks. The second new table will hold Grade and Feedback under AssessmentID.

- Split RESULT into two tables:
 - **RESULT**: {ResultID, StudentID, AssessmentID, ObtainedMarks, TotalMarks}.
 - **GRADE_FEEDBACK**: {AssessmentID, Grade, Feedback}.

The Resulting Tables in 2NF is given below:

1. **STUDENT**: StudentID (PK), FirstName, LastName, Email, EnrollmentDate, ProgramID (FK)
2. **PROGRAM**: ProgramID (PK), ProgramName, Description, Duration, TotalCredits
3. **MODULE**: ModuleID (PK), ModuleName, Description, Credits, Semester, ProgramID (FK)
4. **TEACHER**: TeacherID (PK), FirstName, LastName, Email, Specialization
5. **RESOURCE**: ResourceID (PK), Title, Type, Duration, SequenceNumber, ModuleID (FK)
6. **ANNOUNCEMENT**: AnnouncementID (PK), Title, Content, PostDate, ExpiryDate, ModuleID (FK)
7. **ASSESSMENT**: AssessmentID (PK), Title, Deadline, Weightage, MaximumMarks, ModuleID (FK)
8. **RESULT**: ResultID (PK), AssessmentID (FK), StudentID (FK), ObtainedMarks, TotalMarks

CC5051NT Databases

9. **GRADE_FEEDBACK**: AssessmentID (PK), Grade, Feedback
10. **Linking Tables**:
 - o **Program_Module**: ProgramID (FK), ModuleID (FK)
 - o **Module_Teacher**: ModuleID (FK), TeacherID (FK)

This organized structure helps maintain the integrity of the data and prevents redundancy. By ensuring that all non-key attributes depend fully on the primary key, 2NF enhances the overall efficiency of the database.

3.4. 3NF (Third Normal Form)

Characteristics of 3NF:

Third Normal Form (3NF) has specific characteristics that must be met for a database table to qualify. First and foremost, the table must already be in Second Normal Form (2NF). Being in 2NF means that all non-key attributes should depend on the entire primary key and not just a part of it. This is a crucial step in ensuring data integrity within the database.

An essential part of reaching 3NF involves eliminating transitive dependencies. This means that non-key attributes must not depend on other non-key attributes. A transitive dependency occurs when one non-key attribute relies on another non-key attribute instead of relying solely on the primary key. To avoid this, any non-key attributes that do not depend directly on the primary key must be separated out into their own tables.

In short:-

1. The table must be in 2NF.
2. Eliminate **Transitive Dependencies**: Non-key attributes must not depend on other non-key attributes.

CC5051NT Databases

Transitive Dependency Analysis:

In the analysis of transitive dependencies, after the removal of certain attributes, such as Grade and Feedback, to a separate table, it was confirmed that no transitive dependencies exist. This step is vital in simplifying the design of the database. Each remaining non-key attribute in the tables now strictly depends only on the primary key. This approach helps in organizing the data efficiently and keeps it clean without unnecessary relationships.

In short:-

1. **RESULT:** No transitive dependency exists after removing Grade and Feedback to a separate table.
2. **Other Tables:** Each non-key attribute depends only on the primary key.

The resulting tables that are organized in 3NF include several distinct entities. The STUDENT table comprises fields such as StudentID, FirstName, LastName, Email, EnrollmentDate, and ProgramID, where StudentID serves as the primary key (PK) and ProgramID acts as a foreign key (FK). The PROGRAM table is structured with ProgramID as the primary key and also includes ProgramName, Description, Duration, and TotalCredits.

The MODULE table includes ModuleID as the primary key, along with ModuleName, Description, Credits, Semester, and ProgramID as a foreign key to link it back to the PROGRAM. The TEACHER table contains TeacherID as the primary key with FirstName, LastName, Email, and Specialization as additional fields. The RESOURCE table consists of ResourceID, Title, Type, Duration, SequenceNumber, and ModuleID as the foreign key for connection to the MODULE.

CC5051NT Databases

In addition, the ANNOUNCEMENT table holds AnnouncementID as the primary key alongside Title, Content, PostDate, ExpiryDate, and ModuleID as a foreign key. The ASSESSMENT table has AssessmentID as its primary key, with Title, Deadline, Weightage, MaximumMarks, and ModuleID included. The RESULT table captures results with ResultID as the primary key, plus AssessmentID and StudentID as foreign keys, along with ObtainedMarks and TotalMarks.

A critical addition is the GRADE_FEEDBACK table. This table features AssessmentID as the primary key and includes fields for Grade and Feedback to provide additional context to the assessments. To facilitate relationships between different entities, two linking tables are also created: the Program_Module table links ProgramID and ModuleID, while the Module_Teacher table connects ModuleID and TeacherID.

Resulting Tables in 3NF:

1. **STUDENT**: StudentID (PK), FirstName, LastName, Email, EnrollmentDate, ProgramID (FK)
2. **PROGRAM**: ProgramID (PK), ProgramName, Description, Duration, TotalCredits
3. **MODULE**: ModuleID (PK), ModuleName, Description, Credits, Semester, ProgramID (FK)
4. **TEACHER**: TeacherID (PK), FirstName, LastName, Email, Specialization
5. **RESOURCE**: ResourceID (PK), Title, Type, Duration, SequenceNumber, ModuleID (FK)
6. **ANNOUNCEMENT**: AnnouncementID (PK), Title, Content, PostDate, ExpiryDate, ModuleID (FK)

CC5051NT Databases

7. **ASSESSMENT**: AssessmentID (PK), Title, Deadline, Weightage, MaximumMarks, ModuleID (FK)
8. **RESULT**: ResultID (PK), AssessmentID (FK), StudentID (FK), ObtainedMarks, TotalMarks
9. **GRADE_FEEDBACK**: AssessmentID (PK), Grade, Feedback
10. **Linking Tables**:
 - o **Program_Module**: ProgramID (FK), ModuleID (FK)
 - o **Module_Teacher**: ModuleID (FK), TeacherID (FK)

The normalization process has three key stages. The first stage, First Normal Form (1NF), ensures that all data involves atomic values and avoids repeating groups. The second stage, Second Normal Form (2NF), focuses on removing partial dependencies, which was accomplished by separating Grade and Feedback into the new GRADE_FEEDBACK table. Finally, Third Normal Form (3NF) is achieved by eliminating transitive dependencies, ensuring that every non-key attribute relies solely on its primary key. This structured approach guarantees the integrity and efficiency of the database design.

Summary of Normalization Process:

1. **1NF**: Atomic values and no repeating groups.
2. **2NF**: Removed partial dependencies by separating Grade and Feedback into a new table.
3. **3NF**: Eliminated transitive dependencies, ensuring every non-key attribute depends only on the primary key.

4. Data Dictionary

Here's the data dictionary for the normalized database structure based on 3NF:

CC5051NT Databases

4.1. STUDENT

Table 1 Student data dictionary

Attributes	Data Type	Description
StudentID (PK)	INT	Unique identifier for the student.
FirstName	VARCHAR	First name of the student.
LastName	VARCHAR	Last name of the student.
Email	VARCHAR	Email address of the student.
EnrollmentDate	DATE	Date when the student enrolled.
ProgramID (FK)	INT	Foreign key referring to Program.

4.2. PROGRAM

Table 2 Program data dictionary

Attribute	Data Type	Description
ProgramID (PK)	INT	Unique identifier for the program.
ProgramName	VARCHAR	Name of the program.
Description	TEXT	Description of the program.
Duration	INT	Duration of the program in years.
TotalCredits	INT	Total credits required for the program.

CC5051NT Databases

4.3. MODULE

Table 3 Module data dictionary

Attribute	Data Type	Description
ModuleID (PK)	INT	Unique identifier for the module.
ModuleName	VARCHAR	Name of the module.
Description	TEXT	Description of the module.
Credits	INT	Credits assigned to the module.
Semester	VARCHAR	Semester in which the module is taught.
ProgramID (FK)	INT	Foreign key referring to the Program.

4.4. TEACHER

Table 4 Teacher data dictionary

Attribute	Data Type	Description
TeacherID (PK)	INT	Unique identifier for the teacher.
FirstName	VARCHAR	First name of the teacher.
LastName	VARCHAR	Last name of the teacher.
Email	VARCHAR	Email address of the teacher.
Specialization	VARCHAR	Area of expertise of the teacher.

4.5. RESOURCE

Table 5 Resource data dictionary

Attribute	Data Type	Description
ResourceID (PK)	INT	Unique identifier for the resource.
Title	VARCHAR	Title of the resource.
Type	VARCHAR	Type of the resource (e.g., PDF, video).
Duration	INT	Duration of the resource (if applicable).
SequenceNumber	INT	Sequence order of the resource within the module.
ModuleID (FK)	INT	Foreign key referring to the Module.

4.6. ANNOUNCEMENT

Table 6 Announcement data dictionary

Attribute	Data Type	Description
AnnouncementID (PK)	INT	Unique identifier for the announcement.
Title	VARCHAR	Title of the announcement.
Content	TEXT	Content of the announcement.

CC5051NT Databases

PostDate	DATE	Date the announcement was posted.
ExpiryDate	DATE	Date the announcement expires.
ModuleID (FK)	INT	Foreign key referring to the Module.

4.7. ASSESSMENT

Table 7 Assessment data dictionary

Attribute	Data Type	Description
AssessmentID (PK)	INT	Unique identifier for the assessment.
Title	VARCHAR	Title of the assessment.
Deadline	DATE	Deadline for the assessment submission.
Weightage	INT	Weightage of the assessment.
MaximumMarks	INT	Maximum marks for the assessment.
ModuleID (FK)	INT	Foreign key referring to the Module.

4.8. RESULT

Table 8 Result data dictionary

Attribute	Data Type	Description
-----------	-----------	-------------

CC5051NT Databases

ResultID (PK)	INT	Unique identifier for the result.
AssessmentID (FK)	INT	Foreign key referring to the Assessment.
StudentID (FK)	INT	Foreign key referring to the Student.
ObtainedMarks	INT	Marks obtained by the student.
TotalMarks	INT	Total marks for the assessment.

4.9. GRADE_FEEDBACK

Table 9 Grade_feedback data dictionary

Attribute	Data Type	Description
AssessmentID (PK)	INT	Unique identifier for the assessment.
Grade	VARCHAR	Grade awarded for the assessment.
Feedback	TEXT	Feedback for the student's performance.

4.10. Program_Module (Linking Table)

Table 10 Program_Module data dictionary

Attribute	Data Type	Description
ProgramID (FK)	INT	Foreign key referring to Program.
ModuleID (FK)	INT	Foreign key referring to Module.

CC5051NT Databases

4.11. Module_Teacher (Linking Table)

Table 11 Module_Teacher data dictionary

Attribute	Data Type	Description
ModuleID (FK)	INT	Foreign key referring to Module.
TeacherID (FK)	INT	Foreign key referring to Teacher.

5. Final ERD (Entity-Relationship Diagram)

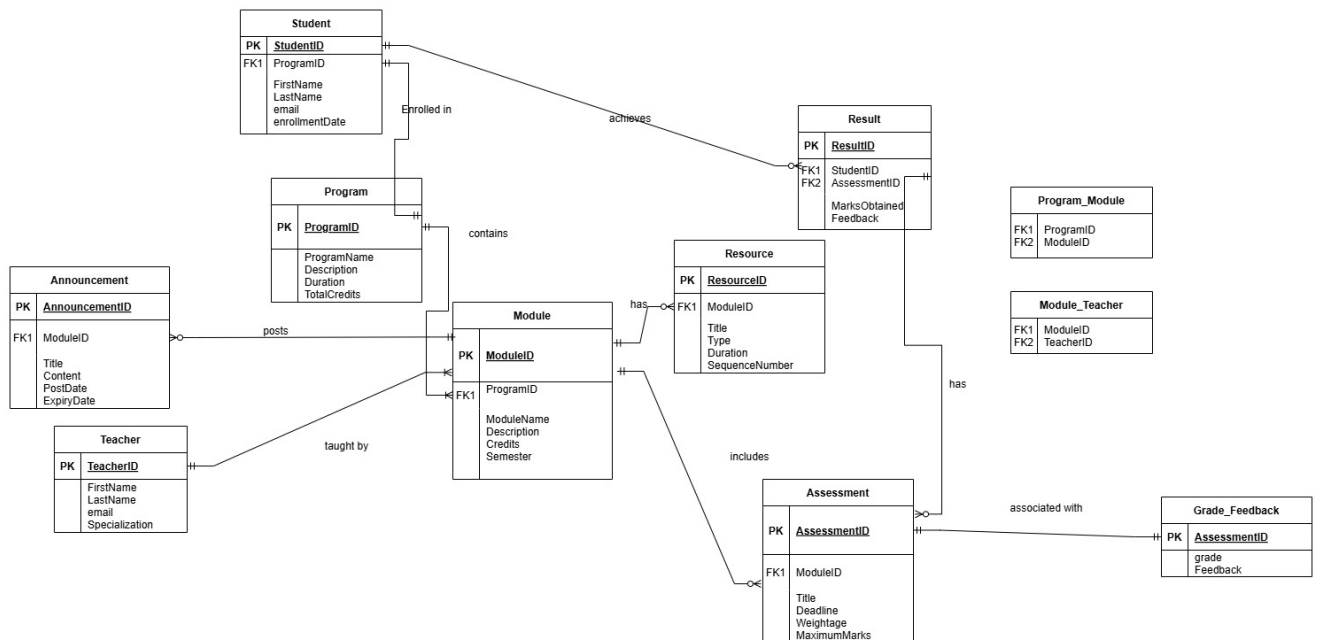


Figure 4 Final ERD

6. Implementation

```
SQL> CREATE TABLE Program (  
2     ProgramID INT PRIMARY KEY,  
3     ProgramName VARCHAR(100),  
4     Description CLOB,  
5     Duration INT,  
6     TotalCredits INT  
7 );
```

Table created.

Figure 5 Program table created

```
SQL> CREATE TABLE Student (  
2     StudentID INT PRIMARY KEY,  
3     FirstName VARCHAR(50),  
4     LastName VARCHAR(50),  
5     Email VARCHAR(100),  
6     EnrollmentDate DATE,  
7     ProgramID INT,  
8     FOREIGN KEY (ProgramID) REFERENCES Program(ProgramID)  
9 );
```

Table created.

Figure 6 Student table created

CC5051NT Databases

```
SQL> CREATE TABLE Module (  
2     ModuleID INT PRIMARY KEY,  
3     ModuleName VARCHAR(100),  
4     Description CLOB,  
5     Credits INT,  
6     Semester VARCHAR(20),  
7     ProgramID INT,  
8     FOREIGN KEY (ProgramID) REFERENCES Program(ProgramID)  
9 );
```

Table created.

Figure 7 Module table created

```
SQL> CREATE TABLE Teacher (  
2     TeacherID INT PRIMARY KEY,  
3     FirstName VARCHAR(50),  
4     LastName VARCHAR(50),  
5     Email VARCHAR(100),  
6     Specialization VARCHAR(100)  
7 );
```

Table created.

Figure 8 Teacher Table created

```
SQL> CREATE TABLE "Resource" (  
2     ResourceID INT PRIMARY KEY,  
3     Title VARCHAR(100),  
4     Type VARCHAR(50),  
5     Duration INT,  
6     SequenceNumber INT,  
7     ModuleID INT,  
8     FOREIGN KEY (ModuleID) REFERENCES Module(ModuleID)  
9 );
```

Table created.

Figure 9 Resource table created

```
SQL> CREATE TABLE Announcement (  
2     AnnouncementID INT PRIMARY KEY,  
3     Title VARCHAR(100),  
4     Content CLOB,  
5     PostDate DATE,  
6     ExpiryDate DATE,  
7     ModuleID INT,  
8     FOREIGN KEY (ModuleID) REFERENCES Module(ModuleID)  
9 );
```

Table created.

Figure 10 Announcement table created

```
SQL> CREATE TABLE Assessment (  
2     AssessmentID INT PRIMARY KEY,  
3     Title VARCHAR(100),  
4     Deadline DATE,  
5     Weightage INT,  
6     MaximumMarks INT,  
7     ModuleID INT,  
8     FOREIGN KEY (ModuleID) REFERENCES Module(ModuleID)  
9 );
```

Table created.

Figure 11 Assessment table created

CC5051NT Databases

```
SQL> CREATE TABLE Result (  
2     ResultID INT PRIMARY KEY,  
3     AssessmentID INT,  
4     StudentID INT,  
5     ObtainedMarks INT,  
6     TotalMarks INT,  
7     FOREIGN KEY (AssessmentID) REFERENCES Assessment(AssessmentID),  
8     FOREIGN KEY (StudentID) REFERENCES Student(StudentID)  
9 );  
  
Table created.
```

Figure 12 Result table created

```
SQL> CREATE TABLE Grade_Feedback (  
2     AssessmentID INT PRIMARY KEY,  
3     Grade VARCHAR(10),  
4     Feedback CLOB,  
5     FOREIGN KEY (AssessmentID) REFERENCES Assessment(AssessmentID)  
6 );  
  
Table created.
```

Figure 13 Grade_Feedback table created

```
SQL> CREATE TABLE Program_Module (  
2     ProgramID INT,  
3     ModuleID INT,  
4     PRIMARY KEY (ProgramID, ModuleID),  
5     FOREIGN KEY (ProgramID) REFERENCES Program(ProgramID),  
6     FOREIGN KEY (ModuleID) REFERENCES Module(ModuleID)  
7 );  
  
Table created.
```

Figure 14 Program_Module table created

CC5051NT Databases

```
SQL> CREATE TABLE Module_Teacher (  
2     ModuleID INT,  
3     TeacherID INT,  
4     PRIMARY KEY (ModuleID, TeacherID),  
5     FOREIGN KEY (ModuleID) REFERENCES Module(ModuleID),  
6     FOREIGN KEY (TeacherID) REFERENCES Teacher(TeacherID)  
7 );  
  
Table created.
```

Figure 15 Module_Teacher table created

```
SQL> INSERT ALL  
2     INTO Program (ProgramID, ProgramName, Description, Duration, TotalCredits)  
3     VALUES (1, 'Computer Science', 'Focuses on computer systems and software development', 4, 120)  
4     INTO Program (ProgramID, ProgramName, Description, Duration, TotalCredits)  
5     VALUES (2, 'Business Administration', 'Covers management and business operations', 3, 90)  
6     INTO Program (ProgramID, ProgramName, Description, Duration, TotalCredits)  
7     VALUES (3, 'Engineering', 'Comprehensive engineering principles and practices', 4, 130)  
8     INTO Program (ProgramID, ProgramName, Description, Duration, TotalCredits)  
9     VALUES (4, 'Psychology', 'Study of human mind and behavior', 3, 100)  
10    INTO Program (ProgramID, ProgramName, Description, Duration, TotalCredits)  
11    VALUES (5, 'Mathematics', 'Explores advanced mathematics and applications', 3, 95)  
12    INTO Program (ProgramID, ProgramName, Description, Duration, TotalCredits)  
13    VALUES (6, 'Biology', 'Study of living organisms', 3, 110)  
14    INTO Program (ProgramID, ProgramName, Description, Duration, TotalCredits)  
15    VALUES (7, 'Physics', 'Comprehensive physics concepts', 4, 125)  
16    SELECT * FROM dual;  
  
7 rows created.
```

Figure 16 Detail inserted in Program

```
SQL> INSERT ALL  
2     INTO Student (StudentID, FirstName, LastName, Email, EnrollmentDate, ProgramID)  
3     VALUES (1, 'Ram', 'Karki', 'ramkarki01@gmail.com', TO_DATE('2023-09-01', 'YYYY-MM-DD'), 1)  
4     INTO Student (StudentID, FirstName, LastName, Email, EnrollmentDate, ProgramID)  
5     VALUES (2, 'Shyam', 'Thapa', 'shyamthapa02@gmail.com', TO_DATE('2023-01-15', 'YYYY-MM-DD'), 2)  
6     INTO Student (StudentID, FirstName, LastName, Email, EnrollmentDate, ProgramID)  
7     VALUES (3, 'Sita', 'Karki', 'sitakarki02@gmail.com', TO_DATE('2022-08-20', 'YYYY-MM-DD'), 3)  
8     INTO Student (StudentID, FirstName, LastName, Email, EnrollmentDate, ProgramID)  
9     VALUES (4, 'Gita', 'Thapa', 'gitathapa04@gmail.com', TO_DATE('2023-06-10', 'YYYY-MM-DD'), 4)  
10    INTO Student (StudentID, FirstName, LastName, Email, EnrollmentDate, ProgramID)  
11    VALUES (5, 'Hari', 'Khadka', 'harikhadka05@gmail.com', TO_DATE('2022-10-01', 'YYYY-MM-DD'), 5)  
12    INTO Student (StudentID, FirstName, LastName, Email, EnrollmentDate, ProgramID)  
13    VALUES (6, 'Laxmi', 'Thapa', 'laxmithapa06@gmail.com', TO_DATE('2023-07-01', 'YYYY-MM-DD'), 6)  
14    INTO Student (StudentID, FirstName, LastName, Email, EnrollmentDate, ProgramID)  
15    VALUES (7, 'Krishna', 'Khatri', 'krishnakhatri07@gmail.com', TO_DATE('2023-09-15', 'YYYY-MM-DD'), 7)  
16    SELECT * FROM dual;  
  
7 rows created.
```

Figure 17 Detail inserted in Student

CC5051NT Databases

```
SQL> INSERT ALL
 2  INTO Module (ModuleID, ModuleName, Description, Credits, Semester, ProgramID)
 3  VALUES (1, 'Databases', 'Covers relational databases and SQL', 20, 'Fall', 1)
 4  INTO Module (ModuleID, ModuleName, Description, Credits, Semester, ProgramID)
 5  VALUES (2, 'Business Strategy', 'Focuses on business planning and decision-making', 15, 'Fall', 2)
 6  INTO Module (ModuleID, ModuleName, Description, Credits, Semester, ProgramID)
 7  VALUES (3, 'Thermodynamics', 'Study of heat and energy transfer', 20, 'Spring', 3)
 8  INTO Module (ModuleID, ModuleName, Description, Credits, Semester, ProgramID)
 9  VALUES (4, 'Cognitive Psychology', 'Explores mental processes and behaviors', 15, 'Spring', 4)
10  INTO Module (ModuleID, ModuleName, Description, Credits, Semester, ProgramID)
11  VALUES (5, 'Linear Algebra', 'Advanced matrix computations and applications', 20, 'Fall', 5)
12  INTO Module (ModuleID, ModuleName, Description, Credits, Semester, ProgramID)
13  VALUES (6, 'Genetics', 'Introduction to genetic principles and analysis', 20, 'Spring', 6)
14  INTO Module (ModuleID, ModuleName, Description, Credits, Semester, ProgramID)
15  VALUES (7, 'Quantum Physics', 'Basic quantum mechanics and applications', 20, 'Fall', 7)
16  SELECT * FROM dual;

7 rows created.
```

Figure 18 Detail inserted in Module

```
SQL> INSERT ALL
 2  INTO Teacher (TeacherID, FirstName, LastName, Email, Specialization)
 3  VALUES (1, 'Animesh', 'Karki', 'animeshkarki01@gmail.com', 'Relational Databases')
 4  INTO Teacher (TeacherID, FirstName, LastName, Email, Specialization)
 5  VALUES (2, 'Dan', 'Karki', 'dankarki02@gmail.com', 'Business Strategy')
 6  INTO Teacher (TeacherID, FirstName, LastName, Email, Specialization)
 7  VALUES (3, 'Ambika', 'Basnet', 'ambikabasnet03@gmail.com', 'Thermodynamics')
 8  INTO Teacher (TeacherID, FirstName, LastName, Email, Specialization)
 9  VALUES (4, 'Kalpana', 'Thapa', 'kalpanathapa04@gmail.com', 'Cognitive Psychology')
10  INTO Teacher (TeacherID, FirstName, LastName, Email, Specialization)
11  VALUES (5, 'Anusca', 'Karki', 'anuscakarki05@gmail.com', 'Linear Algebra')
12  INTO Teacher (TeacherID, FirstName, LastName, Email, Specialization)
13  VALUES (6, 'Anusha', 'Basnet', 'anushabasnet06@gmail.com', 'Genetics')
14  INTO Teacher (TeacherID, FirstName, LastName, Email, Specialization)
15  VALUES (7, 'Sandip', 'Khatrri', 'sandipkhatrri07@gmail.com', 'Quantum Physics')
16  SELECT * FROM dual;

7 rows created.
```

Figure 19 Detail inserted in Teacher

CC5051NT Databases

```
SQL> INSERT ALL
  2   INTO "Resource" (ResourceID, Title, Type, Duration, SequenceNumber, ModuleID)
  3   VALUES (1, 'Lecture: Intro to Databases', 'Video', 60, 1, 1)
  4   INTO "Resource" (ResourceID, Title, Type, Duration, SequenceNumber, ModuleID)
  5   VALUES (2, 'Lecture: Advanced SQL', 'Video', 75, 2, 1)
  6   INTO "Resource" (ResourceID, Title, Type, Duration, SequenceNumber, ModuleID)
  7   VALUES (3, 'Case Study: Business Planning', 'PDF', NULL, 1, 2)
  8   INTO "Resource" (ResourceID, Title, Type, Duration, SequenceNumber, ModuleID)
  9   VALUES (4, 'Lecture: Laws of Thermodynamics', 'Video', 90, 1, 3)
 10  INTO "Resource" (ResourceID, Title, Type, Duration, SequenceNumber, ModuleID)
 11  VALUES (5, 'Lab Manual: Heat Transfer', 'PDF', NULL, 2, 3)
 12  INTO "Resource" (ResourceID, Title, Type, Duration, SequenceNumber, ModuleID)
 13  VALUES (6, 'Lecture: Cognitive Models', 'Video', 55, 1, 4)
 14  INTO "Resource" (ResourceID, Title, Type, Duration, SequenceNumber, ModuleID)
 15  VALUES (7, 'Lab Guide: Matrices and Applications', 'PDF', NULL, 1, 5)

 16  SELECT * FROM dual;

7 rows created.
```

Figure 20 Detail inserted in "Resource"

```
SQL> INSERT ALL
  2   INTO Announcement (AnnouncementID, Title, Content, PostDate, ExpiryDate, ModuleID)
  3   VALUES (1, 'Midterm Exam Notice', 'Midterm exams scheduled for next month.', TO_DATE('2024-04-15', 'YYYY-MM-DD'), TO_DATE('2024-05-15', 'YYYY-MM-DD')
  4   , 1)
  5   INTO Announcement (AnnouncementID, Title, Content, PostDate, ExpiryDate, ModuleID)
  6   VALUES (2, 'Project Submission Guidelines', 'Guidelines for project submissions.', TO_DATE('2024-04-20', 'YYYY-MM-DD'), TO_DATE('2024-05-25', 'YYYY-MM-DD'), 2)
  7   INTO Announcement (AnnouncementID, Title, Content, PostDate, ExpiryDate, ModuleID)
  8   VALUES (3, 'Lab Session Update', 'Lab sessions will be rescheduled.', TO_DATE('2024-04-25', 'YYYY-MM-DD'), TO_DATE('2024-05-30', 'YYYY-MM-DD'), 3)
  9   INTO Announcement (AnnouncementID, Title, Content, PostDate, ExpiryDate, ModuleID)
 10   VALUES (4, 'New Lecture Uploaded', 'Lecture on Cognitive Models uploaded.', TO_DATE('2024-04-18', 'YYYY-MM-DD'), TO_DATE('2024-05-15', 'YYYY-MM-DD'), 4)
 11   INTO Announcement (AnnouncementID, Title, Content, PostDate, ExpiryDate, ModuleID)
 12   VALUES (5, 'Assignment Reminder', 'Submit your assignments before the deadline.', TO_DATE('2024-04-10', 'YYYY-MM-DD'), TO_DATE('2024-05-01', 'YYYY-MM-DD'), 5)
 13   INTO Announcement (AnnouncementID, Title, Content, PostDate, ExpiryDate, ModuleID)
 14   VALUES (6, 'Workshop Announcement', 'Workshop on advanced genetics.', TO_DATE('2024-04-22', 'YYYY-MM-DD'), TO_DATE('2024-05-20', 'YYYY-MM-DD'), 6)
 15   INTO Announcement (AnnouncementID, Title, Content, PostDate, ExpiryDate, ModuleID)
 16   VALUES (7, 'Seminar Registration', 'Register for the Quantum Physics seminar.', TO_DATE('2024-04-05', 'YYYY-MM-DD'), TO_DATE('2024-05-10', 'YYYY-MM-DD'), 7)
 17  SELECT * FROM dual;

7 rows created.
```

Figure 21 Detail inserted in Announcement

CC5051NT Databases

```
SQL> INSERT ALL
 2  INTO Assessment (AssessmentID, Title, Deadline, Weightage, MaximumMarks, ModuleID)
 3  VALUES (1, 'Quiz 1', TO_DATE('2024-05-01', 'YYYY-MM-DD'), 10, 20, 1)
 4  INTO Assessment (AssessmentID, Title, Deadline, Weightage, MaximumMarks, ModuleID)
 5  VALUES (2, 'Midterm Exam', TO_DATE('2024-05-15', 'YYYY-MM-DD'), 30, 50, 1)
 6  INTO Assessment (AssessmentID, Title, Deadline, Weightage, MaximumMarks, ModuleID)
 7  VALUES (3, 'Business Case Analysis', TO_DATE('2024-05-20', 'YYYY-MM-DD'), 20, 40, 2)
 8  INTO Assessment (AssessmentID, Title, Deadline, Weightage, MaximumMarks, ModuleID)
 9  VALUES (4, 'Thermodynamics Lab Report', TO_DATE('2024-05-25', 'YYYY-MM-DD'), 15, 30, 3)
10  INTO Assessment (AssessmentID, Title, Deadline, Weightage, MaximumMarks, ModuleID)
11  VALUES (5, 'Cognitive Research Paper', TO_DATE('2024-05-30', 'YYYY-MM-DD'), 25, 50, 4)
12  INTO Assessment (AssessmentID, Title, Deadline, Weightage, MaximumMarks, ModuleID)
13  VALUES (6, 'Matrix Applications Assignment', TO_DATE('2024-06-01', 'YYYY-MM-DD'), 20, 40, 5)
14  INTO Assessment (AssessmentID, Title, Deadline, Weightage, MaximumMarks, ModuleID)
15  VALUES (7, 'Quantum Mechanics Quiz', TO_DATE('2024-06-05', 'YYYY-MM-DD'), 10, 20, 7)
16  SELECT * FROM dual;

7 rows created.
```

Figure 22 Detail inserted in Assessment

```
SQL> INSERT ALL
 2  INTO Result (ResultID, AssessmentID, StudentID, ObtainedMarks, TotalMarks)
 3  VALUES (1, 1, 1, 18, 20)
 4  INTO Result (ResultID, AssessmentID, StudentID, ObtainedMarks, TotalMarks)
 5  VALUES (2, 2, 1, 40, 50)
 6  INTO Result (ResultID, AssessmentID, StudentID, ObtainedMarks, TotalMarks)
 7  VALUES (3, 1, 2, 15, 20)
 8  INTO Result (ResultID, AssessmentID, StudentID, ObtainedMarks, TotalMarks)
 9  VALUES (4, 2, 2, 35, 50)
10  INTO Result (ResultID, AssessmentID, StudentID, ObtainedMarks, TotalMarks)
11  VALUES (5, 3, 3, 30, 40)
12  INTO Result (ResultID, AssessmentID, StudentID, ObtainedMarks, TotalMarks)
13  VALUES (6, 4, 4, 25, 30)
14  INTO Result (ResultID, AssessmentID, StudentID, ObtainedMarks, TotalMarks)
15  VALUES (7, 5, 5, 45, 50)
16  SELECT * FROM dual;

7 rows created.
```

Figure 23 Detail inserted in Result

CC5051NT Databases

```
SQL> INSERT ALL
  2   INTO Grade_Feedback (AssessmentID, Grade, Feedback)
  3   VALUES (1, 'A', 'Excellent performance.')
  4   INTO Grade_Feedback (AssessmentID, Grade, Feedback)
  5   VALUES (2, 'B+', 'Good effort, but can improve.')
  6   INTO Grade_Feedback (AssessmentID, Grade, Feedback)
  7   VALUES (3, 'A-', 'Well analyzed and presented.')
  8   INTO Grade_Feedback (AssessmentID, Grade, Feedback)
  9   VALUES (4, 'B', 'Good understanding of concepts.')
 10   INTO Grade_Feedback (AssessmentID, Grade, Feedback)
 11   VALUES (5, 'A', 'Exceptional work on research paper.')
 12   INTO Grade_Feedback (AssessmentID, Grade, Feedback)
 13   VALUES (6, 'A-', 'Solid work on matrix assignment.')
 14   INTO Grade_Feedback (AssessmentID, Grade, Feedback)
 15   VALUES (7, 'B+', 'Good understanding of quantum concepts.')
 16   SELECT * FROM dual;

7 rows created.
```

Figure 24 Detail inserted in Grade_Feedback

```
SQL> INSERT ALL
  2   INTO Program_Module (ProgramID, ModuleID) VALUES (1, 1)
  3   INTO Program_Module (ProgramID, ModuleID) VALUES (2, 2)
  4   INTO Program_Module (ProgramID, ModuleID) VALUES (3, 3)
  5   INTO Program_Module (ProgramID, ModuleID) VALUES (4, 4)
  6   INTO Program_Module (ProgramID, ModuleID) VALUES (5, 5)
  7   INTO Program_Module (ProgramID, ModuleID) VALUES (6, 6)
  8   INTO Program_Module (ProgramID, ModuleID) VALUES (7, 7)
  9   SELECT * FROM dual;

7 rows created.
```

Figure 25 Detail inserted in Program_Module

```
SQL> INSERT ALL
  2   INTO Module_Teacher (ModuleID, TeacherID) VALUES (1, 1)
  3   INTO Module_Teacher (ModuleID, TeacherID) VALUES (2, 1)
  4   INTO Module_Teacher (ModuleID, TeacherID) VALUES (3, 2)
  5   INTO Module_Teacher (ModuleID, TeacherID) VALUES (4, 2)
  6   INTO Module_Teacher (ModuleID, TeacherID) VALUES (5, 3)
  7   INTO Module_Teacher (ModuleID, TeacherID) VALUES (6, 4)
  8   INTO Module_Teacher (ModuleID, TeacherID) VALUES (7, 7)
  9   SELECT * FROM dual;

7 rows created.
```

Figure 26 Detail inserted in Module_Teacher

CC5051NT Databases

```
SQL> SELECT * FROM Program;

PROGRAMID
PROGRAMNAME
DESCRIPTION
DURATION TOTALCREDITS
1
Computer Science
Focuses on computer systems and software development
4 128

PROGRAMID
PROGRAMNAME
DESCRIPTION
DURATION TOTALCREDITS
2
Business Administration
Covers management and business operations
3 90

PROGRAMID
PROGRAMNAME
DESCRIPTION
DURATION TOTALCREDITS
3
Engineering
Comprehensive engineering principles and practices
4 130

PROGRAMID
PROGRAMNAME
DESCRIPTION
DURATION TOTALCREDITS
4
Psychology
Study of human mind and behavior
3 100

PROGRAMID
PROGRAMNAME
DESCRIPTION
DURATION TOTALCREDITS
5
Mathematics
Explores advanced mathematics and applications
3 95

PROGRAMID
PROGRAMNAME
DESCRIPTION
DURATION TOTALCREDITS
6
Biology
Study of living organisms
3 110

PROGRAMID
PROGRAMNAME
DESCRIPTION
DURATION TOTALCREDITS
7
Physics
Comprehensive physics concepts
4 125

7 rows selected.
```

Figure 27 Details of Program

CC5051NT Databases

```
SQL> SELECT * FROM Student;

STUDENTID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
ENROLLMENT PROGRAMID
-----
1 Ran
Mark1
mark101@gmail.com
01-SEP-23 1

STUDENTID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
ENROLLMENT PROGRAMID
-----
2 Shyan
Thapa
shyanthapa02@gmail.com
15-JAN-23 2

STUDENTID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
ENROLLMENT PROGRAMID
-----
3 Sita
Mark1
sitakark102@gmail.com
20-AUG-22 3

STUDENTID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
ENROLLMENT PROGRAMID
-----
4 Gita
Thapa
gitathapa04@gmail.com
10-JUN-23 4

STUDENTID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
ENROLLMENT PROGRAMID
-----
5 Hari
Khodja
harikhodja05@gmail.com
01-OCT-22 5

STUDENTID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
ENROLLMENT PROGRAMID
-----
6 Laxmi
Thapa
laxmithapa06@gmail.com
01-JUL-23 6

STUDENTID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
ENROLLMENT PROGRAMID
-----
7 Krishna
Khatr1
krishnakhatr107@gmail.com
15-SEP-23 7

7 rows selected.
```

Figure 28 Details of Student

CC5051NT Databases

```
SQL> SELECT * FROM Module;
```

MODULEID	MODULENAME	DESCRIPTION	CREDITS	SEMESTER	PROGRAMID
1	Databases	Covers relational databases and SQL	20	Fall	1
2	Business Strategy	Focuses on business planning and decision-making	15	Fall	2
3	Thermodynamics	Study of heat and energy transfer	20	Spring	3
4	Cognitive Psychology	Explores mental processes and behaviors	15	Spring	4
5	Linear Algebra	Advanced matrix computations and applications	20	Fall	5
6	Genetics	Introduction to genetic principles and analysis	20	Spring	6
7	Quantum Physics	Basic quantum mechanics and applications	20	Fall	7

7 rows selected.

Figure 29 Details of Module

CC5051NT Databases

```
SQL> SELECT * FROM Teacher;

TEACHERID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
SPECIALIZATION
-----
1 Animesh
Karki
animeshkarki81@gmail.com
Relational Databases

TEACHERID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
SPECIALIZATION
-----
2 Dan
Karki
dankarki82@gmail.com
Business Strategy

TEACHERID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
SPECIALIZATION
-----
3 Ambika
Basnet
ambikabasnet83@gmail.com
Thermodynamics

TEACHERID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
SPECIALIZATION
-----
4 Kalpana
Thapa
kalpanathapa84@gmail.com
Cognitive Psychology

TEACHERID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
SPECIALIZATION
-----
5 Anusca
Karki
anuscakarki85@gmail.com
Linear Algebra

TEACHERID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
SPECIALIZATION
-----
6 Anusha
Basnet
anushabasnet86@gmail.com
Genetics

TEACHERID FIRSTNAME
-----
LASTNAME
-----
EMAIL
-----
SPECIALIZATION
-----
7 Sandip
Khatr
sandipkhatr87@gmail.com
Quantum Physics

7 rows selected.
```

Figure 30 Details of Teacher

CC5051NT Databases

```
SQL> SELECT * FROM "Resource";
```

RESOURCEID	TITLE	TYPE	DURATION	SEQUENCENUMBER
1	Lecture: Intro to Databases	Video	60	1
2	Lecture: Advanced SQL	Video	75	2
3	Case Study: Business Planning	PDF		1
4	Lecture: Laws of Thermodynamics	Video	90	1
5	Lab Manual: Heat Transfer	PDF		2
6	Lecture: Cognitive Models	Video	55	1
7	Lab Guide: Matrices and Applications	PDF		1

7 rows selected.

Figure 31 Details of "Resource"

CC5051NT Databases

```
SQL> SELECT * FROM Announcement;
```

ANNOUNCEMENTID	TITLE	CONTENT	POSTDATE	EXPIRYDAT	MODULEID
1	Midterm Exam Notice	Midterm exams scheduled for next month.	01-MAY-24	01-JUN-24	1
2	Project Submission Guidelines	Guidelines for project submissions.	20-APR-24	25-MAY-24	2
3	Lab Session Update	Lab sessions will be rescheduled.	25-APR-24	30-MAY-24	3
4	New Lecture Uploaded	Lecture on Cognitive Models uploaded.	10-APR-24	15-MAY-24	4
5	Assignment Reminder	Submit your assignments before the deadline.	10-APR-24	01-MAY-24	5
6	Workshop Announcement	Workshop on advanced genetics.	22-APR-24	20-MAY-24	6
7	Seminar Registration	Register for the Quantum Physics seminar.	05-APR-24	10-MAY-24	7

7 rows selected.

Figure 32 Details of Announcement

CC5051NT Databases

```
SQL> SELECT * FROM Assessment;
```

ASSESSMENTID

TITLE

DEADLINE WEIGHTAGE MAXIMUMMARKS MODULEID

1
Quiz 1
01-MAY-24 10 20 1

2
Midterm Exam
15-MAY-24 30 50 1

ASSESSMENTID

TITLE

DEADLINE WEIGHTAGE MAXIMUMMARKS MODULEID

3
Business Case Analysis
20-MAY-24 20 40 2

4
Thermodynamics Lab Report

ASSESSMENTID

TITLE

DEADLINE WEIGHTAGE MAXIMUMMARKS MODULEID

25-MAY-24 15 30 3

5
Cognitive Research Paper
30-MAY-24 25 50 4

6

ASSESSMENTID

TITLE

DEADLINE WEIGHTAGE MAXIMUMMARKS MODULEID

Matrix Applications Assignment
01-JUN-24 20 40 5

7
Quantum Mechanics Quiz
05-JUN-24 10 20 7

7 rows selected.

Figure 33 Details of Assessment

CC5051NT Databases

```
SQL> SELECT * FROM Result;
```

RESULTID	ASSESSMENTID	STUDENTID	OBTAINEDMARKS	TOTALMARKS
1	1	1	18	20
2	2	1	40	50
3	1	2	15	20
4	2	2	35	50
5	3	3	30	40
6	4	4	25	30
7	5	5	45	50

7 rows selected.

Figure 34 Details of Result

```
SQL> SELECT * FROM Grade_Feedback;
```

ASSESSMENTID	GRADE	FEEDBACK
1	A	Excellent performance.
2	B+	Good effort, but can improve.
3	A-	Well analyzed and presented.
4	B	Good understanding of concepts.
5	A	Exceptional work on research paper.
6	A-	Solid work on matrix assignment.
7	B+	Good understanding of quantum concepts.

7 rows selected.

Figure 35 Details of Grade_Feedback

CC5051NT Databases

```
SQL> SELECT * FROM Program_Module;
```

PROGRAMID	MODULEID
1	1
2	2
3	3
4	4
5	5
6	6
7	7

7 rows selected.

Figure 36 Details of Program_Module

```
SQL> SELECT * FROM Module_Teacher;
```

MODULEID	TEACHERID
1	1
2	1
3	2
4	2
5	3
6	4
7	7

7 rows selected.

Figure 37 Details of Module_Teacher

7. Database Querying

7.1. Information query

```
SQL> SELECT
  2     Program.ProgramName,
  3     COUNT(Student.StudentID) AS TotalStudents
  4 FROM
  5     Program
  6 LEFT JOIN
  7     Student
  8 ON
  9     Program.ProgramID = Student.ProgramID
 10 GROUP BY
 11     Program.ProgramName;
```

PROGRAMNAME

TOTALSTUDENTS

Mathematics
1

Physics
1

Computer Science
1

PROGRAMNAME

TOTALSTUDENTS

Business Administration
1

Engineering
1

Biology
1

PROGRAMNAME

TOTALSTUDENTS

Psychology
1

7 rows selected.

Figure 38 Program and total no. of student

CC5051NT Databases

```
SQL> SELECT
2     Title, Content, PostDate
3 FROM
4     Announcement
5 WHERE
6     ModuleID = 1
7     AND PostDate BETWEEN TO_DATE('2024-05-01', 'YYYY-MM-DD') AND TO_DATE('2024-05-28', 'YYYY-MM-DD');

TITLE
-----
CONTENT
-----
POSTDATE
-----
Midterm Exam Notice
Midterm exams scheduled for next month.
01-MAY-24
```

Figure 39 Announcement

```
SQL> SELECT
2     Module.ModuleName,
3     COUNT("Resource".ResourceID) AS TotalResources
4 FROM
5     Module
6 LEFT JOIN
7     "Resource"
8 ON
9     Module.ModuleID = "Resource".ModuleID
10 WHERE
11     Module.ModuleName LIKE 'D%'
12 GROUP BY
13     Module.ModuleName;

MODULENAME
-----
TOTALRESOURCES
-----
Databases          2
```

Figure 40 Module with D

CC5051NT Databases

```
SQL> SELECT
  2     Student.FirstName, Student.LastName, Program.ProgramName
  3 FROM
  4     Student
  5 JOIN
  6     Program
  7 ON
  8     Student.ProgramID = Program.ProgramID
  9 LEFT JOIN
 10     Result
 11 ON
 12     Student.StudentID = Result.StudentID
 13 WHERE
 14     Result.AssessmentID IS NULL;
```

FIRSTNAME

LASTNAME

PROGRAMNAME

Laxmi
Thapa
Biology

Krishna
Khatri
Physics

FIRSTNAME

LASTNAME

PROGRAMNAME

Figure 41 not submitted

```
SQL> SELECT
  2     Teacher.FirstName, Teacher.LastName, COUNT(Module_Teacher.ModuleID) AS TotalModules
  3 FROM
  4     Teacher
  5 JOIN
  6     Module_Teacher
  7 ON
  8     Teacher.TeacherID = Module_Teacher.TeacherID
  9 GROUP BY
 10     Teacher.FirstName, Teacher.LastName
 11 HAVING
 12     COUNT(Module_Teacher.ModuleID) > 1;
```

FIRSTNAME

LASTNAME

TOTALMODULES

Animesh
Karki 2

Dan
Karki 2

Figure 42 more than 1 module

7.2. Transaction query

```
SQL> SELECT * FROM (  
  2     SELECT  
  3         Module.ModuleName,  
  4         MAX(Assessment.Deadline) AS LatestDeadline  
  5     FROM  
  6         Module  
  7     JOIN  
  8         Assessment  
  9     ON  
 10         Module.ModuleID = Assessment.ModuleID  
 11     GROUP BY  
 12         Module.ModuleName  
 13     ORDER BY  
 14         LatestDeadline DESC  
 15 )  
 16 WHERE ROWNUM = 1;
```

MODULENAME

LATESTDEA

Quantum Physics
05-JUN-24

Figure 43 Latest assessment deadline

CC5051NT Databases

```
SQL> SELECT * FROM (  
  2     SELECT  
  3         Student.FirstName,  
  4         Student.LastName,  
  5         SUM(Result.ObtainedMarks) AS TotalScore  
  6     FROM  
  7         Student  
  8     JOIN  
  9         Result  
 10     ON  
 11         Student.StudentID = Result.StudentID  
 12     GROUP BY  
 13         Student.FirstName, Student.LastName  
 14     ORDER BY  
 15         TotalScore DESC  
 16 )  
 17 WHERE ROWNUM <= 3;
```

FIRSTNAME	

LASTNAME	TOTALSCORE
-----	-----
Ram	
Karki	58
Shyam	
Thapa	50
Hari	
Khadka	45

Figure 44 Highest score of 3 student

CC5051NT Databases

```
SQL> SELECT
  2     Program.ProgramName,
  3     COUNT(Assessment.AssessmentID) AS TotalAssessments,
  4     AVG(Result.ObtainedMarks) AS AverageScore
  5 FROM
  6     Program
  7 JOIN
  8     Module
  9 ON
 10     Program.ProgramID = Module.ProgramID
 11 JOIN
 12     Assessment
 13 ON
 14     Module.ModuleID = Assessment.ModuleID
 15 JOIN
 16     Result
 17 ON
 18     Assessment.AssessmentID = Result.AssessmentID
 19 GROUP BY
 20     Program.ProgramName;
```

PROGRAMNAME

TOTALASSESSMENTS AVERAGESCORE

Computer Science	4	27
------------------	---	----

Business Administration	1	30
-------------------------	---	----

Engineering	1	25
-------------	---	----

PROGRAMNAME

TOTALASSESSMENTS AVERAGESCORE

Psychology	1	45
------------	---	----

Figure 45 Average in database

CC5051NT Databases

```
SQL> SELECT
  2     Student.FirstName, Student.LastName, Result.ObtainedMarks
  3 FROM
  4     Student
  5 JOIN
  6     Result
  7 ON
  8     Student.StudentID = Result.StudentID
  9 JOIN
 10     Assessment
 11 ON
 12     Result.AssessmentID = Assessment.AssessmentID
 13 JOIN
 14     Module
 15 ON
 16     Assessment.ModuleID = Module.ModuleID
 17 WHERE
 18     Module.ModuleName = 'Databases'
 19     AND Result.ObtainedMarks > (
 20         SELECT AVG(Result.ObtainedMarks)
 21         FROM Result
 22         JOIN Assessment
 23         ON Result.AssessmentID = Assessment.AssessmentID
 24         WHERE ModuleID = Module.ModuleID
 25     );
```

FIRSTNAME	

LASTNAME	OBTAINEDMARKS
-----	-----
Shyam	
Thapa	35
Ram	
Karki	40

Figure 46 total no. and average in assessment

```

SQL> SELECT
  2     Student.FirstName, Student.LastName,
  3     CASE
  4         WHEN SUM(Result.ObtainedMarks) >= 50 THEN 'Pass'
  5         ELSE 'Fail'
  6     END AS Remarks
  7 FROM
  8     Student
  9 JOIN
 10     Result
 11 ON
 12     Student.StudentID = Result.StudentID
 13 JOIN
 14     Assessment
 15 ON
 16     Result.AssessmentID = Assessment.AssessmentID
 17 WHERE
 18     Assessment.ModuleID = 1
 19 GROUP BY
 20     Student.FirstName, Student.LastName;

```

FIRSTNAME	LASTNAME	REMA
Ram	Karki	Pass
Shyam	Thapa	Pass

Figure 47 pass or fail in module

8. Critical Evaluation

8.1. Critical Evaluation of Module

The program successfully gives students the information and abilities they need to evaluate, create, and build database systems. It establishes a solid theoretical foundation by presenting important database design concepts including normalization, data modeling, and entity-relationship diagrams. Students' ability to convert designs into useful databases is further improved by practical

CC5051NT Databases

implementation using Structured Query Language (SQL), which closely resembles real-world applications. Students may adapt database systems to particular use cases while preserving efficiency and referential integrity thanks to the combination of analysis and design approaches. Furthermore, by emphasizing relational database management systems (RDBMS), a commonly used business standard, the module conforms to industry standards.

The module does have certain restrictions, though. In-depth discussions of complex subjects like indexing, triggers, stored procedures, and query optimization—all crucial for handling big datasets and maximizing performance—are absent. Additionally, it fails to expose students to other database technologies, such as NoSQL databases, which are becoming more and more popular for managing vast amounts of unstructured data. Students are also unable to have practical experience with modern database settings because of the module's limited exposure to contemporary technologies and cloud platforms. The program may be improved by including more complex subjects such as database security, indexing, and query optimization in addition to an overview of NoSQL databases and a comparison with traditional databases. Practical problem-solving abilities would be improved by including real-world project-based assignments, and exposure to cloud-based database systems, MySQL Workbench, and Oracle SQL Developer would close the knowledge gap between academic study and business needs. The program would provide students a more thorough education by covering these topics, better equipping them to handle the issues of contemporary databases.

8.2. Critical Assessment of Coursework

The curriculum is designed to help students develop their fundamental understanding of relational databases and their ability to execute them practically. It provides a methodical approach that starts with normalization principles, entity-relationship diagrams (ERD), and the theoretical knowledge of relational databases. It ends with the implementation and querying of a fully working database. By allowing students to relate abstract ideas to real-world applications, this methodical approach promotes a thorough comprehension of database systems.

Emphasizing database normalization is one of the coursework's advantages. The emphasis on unnormalized form (UNF), first normal form (1NF), second normal form (2NF), and third normal form (3NF) guarantees that students will learn how to eliminate redundancy and assure data integrity in order to construct database systems that are scalable and efficient. This is essential for creating systems that can manage the complexity of data in the actual world. Furthermore, the incorporation of referential integrity via primary and foreign keys enhances comprehension of relational database concepts.

Additionally, there are projects for constructing databases using SQL in the curriculum, which is quite relevant and useful. Students have practical experience with SQL for data management through the use of queries like SELECT, JOIN, and aggregate functions, as well as activities like table construction and data insertion. However, by include more complex SQL concepts—like window functions, stored procedures, and triggers—which are becoming more and more

CC5051NT Databases

necessary in contemporary database systems, the course might be improved even further.

The activities' practical relevance is another area that needs work. The examples and datasets offered are instructive, but adding additional business-oriented situations would better equip students to face the problems of the industry. Tasks like managing intricate relationships, optimizing big datasets, and performance tweaking, for example, might be quite beneficial.

Additionally, the course does not introduce students to other database paradigms, such as NoSQL databases, which are becoming more popular due to their ability to handle unstructured data and scalability issues. Relational databases and these ideas introduced together would provide database administration a more comprehensive grasp.

Regarding the design, implementation, and querying of relational databases, the course does a good job of explaining the basics. It may provide an even more comprehensive and industry-relevant learning experience by tackling the aforementioned areas for development.

9. Drop Query and Database Dump file creation

```
SQL> DROP TABLE Module_Teacher CASCADE CONSTRAINTS;
Table dropped.

SQL> DROP TABLE Program_Module CASCADE CONSTRAINTS;
Table dropped.

SQL> DROP TABLE Grade_Feedback CASCADE CONSTRAINTS;
Table dropped.

SQL> DROP TABLE Result CASCADE CONSTRAINTS;
Table dropped.

SQL> DROP TABLE Assessment CASCADE CONSTRAINTS;
Table dropped.

SQL> DROP TABLE Announcement CASCADE CONSTRAINTS;
Table dropped.

SQL> DROP TABLE "Resource" CASCADE CONSTRAINTS;
Table dropped.

SQL> DROP TABLE Module CASCADE CONSTRAINTS;
Table dropped.

SQL> DROP TABLE Student CASCADE CONSTRAINTS;
Table dropped.

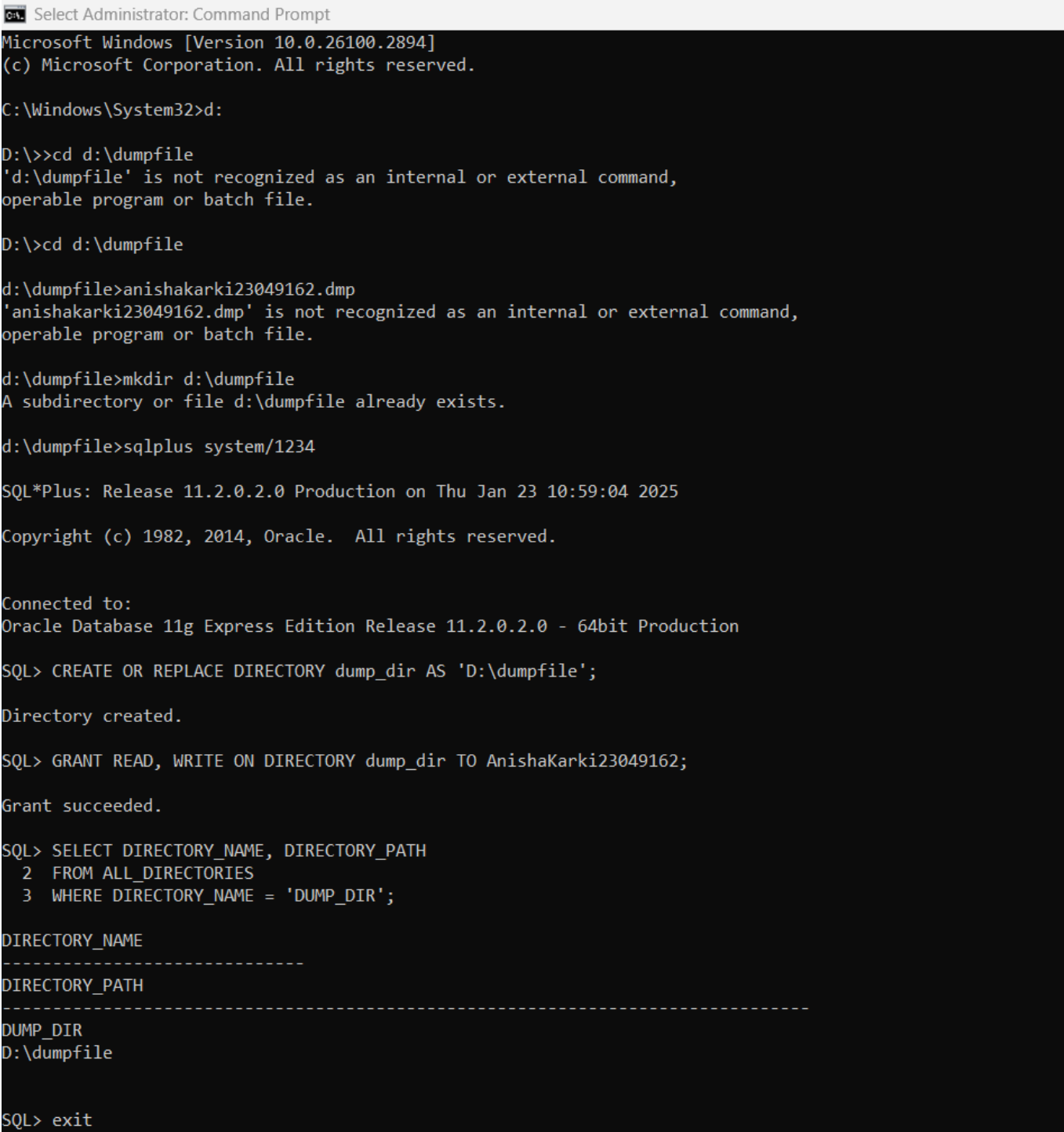
SQL> DROP TABLE Teacher CASCADE CONSTRAINTS;
Table dropped.

SQL> DROP TABLE Program CASCADE CONSTRAINTS;
Table dropped.

SQL> |
```

CC5051NT Databases

Figure 48 Drop Query



```

C:\Windows\System32>cmd
Microsoft Windows [Version 10.0.26100.2894]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>d:

D:\>>cd d:\dumpfile
'd:\dumpfile' is not recognized as an internal or external command,
operable program or batch file.

D:\>cd d:\dumpfile

d:\dumpfile>anishakarki23049162.dmp
'anishakarki23049162.dmp' is not recognized as an internal or external command,
operable program or batch file.

d:\dumpfile>mkdir d:\dumpfile
A subdirectory or file d:\dumpfile already exists.

d:\dumpfile>sqlplus system/1234

SQL*Plus: Release 11.2.0.2.0 Production on Thu Jan 23 10:59:04 2025

Copyright (c) 1982, 2014, Oracle. All rights reserved.

Connected to:
Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production

SQL> CREATE OR REPLACE DIRECTORY dump_dir AS 'D:\dumpfile';

Directory created.

SQL> GRANT READ, WRITE ON DIRECTORY dump_dir TO AnishaKarki23049162;

Grant succeeded.

SQL> SELECT DIRECTORY_NAME, DIRECTORY_PATH
2 FROM ALL_DIRECTORIES
3 WHERE DIRECTORY_NAME = 'DUMP_DIR';

DIRECTORY_NAME
-----
DIRECTORY_PATH
-----
DUMP_DIR
D:\dumpfile

SQL> exit
```

Figure 49 Data dump file creation

CC5051NT Databases

```
SQL> exit
Disconnected from Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production

d:\dumpfile>expdp AnishaKarki23049162/1234 DIRECTORY=dump_dir DUMPFILE=anishakarki23049162.dmp LOGFILE=export.log

Export: Release 11.2.0.2.0 - Production on Thu Jan 23 11:01:03 2025

Copyright (c) 1982, 2009, Oracle and/or its affiliates. All rights reserved.

Connected to: Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production
Starting "ANISHAKARKI23049162"."SYS_EXPORT_SCHEMA_01": AnishaKarki23049162/***** DIRECTORY=dump_dir DUMPFILE=anishakarki23049162.dmp LOGFILE=export.log
Estimate in progress using BLOCKS method...
Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA
Total estimation using BLOCKS method: 0 KB
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/COMMENT
Master table "ANISHAKARKI23049162"."SYS_EXPORT_SCHEMA_01" successfully loaded/unloaded
*****
Dump file set for ANISHAKARKI23049162.SYS_EXPORT_SCHEMA_01 is:
D:\DUMPFILE\ANISHAKARKI23049162.DMP
Job "ANISHAKARKI23049162"."SYS_EXPORT_SCHEMA_01" successfully completed at 11:01:47

d:\dumpfile>
```

Figure 50 Expdp Anishakarki23049162